

EXP 4: Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets. (Deepan Kumar P 192312498)

CODE:

```
# Import libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score, classification_report

# Load dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

# Normalize data

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

# Create ANN model

model = MLPClassifier(

    hidden_layer_sizes=(10,),

    activation='relu',

    solver='adam',

    max_iter=1000,

    random_state=42

)

# Train model (Backpropagation)

model.fit(X_train, y_train)
```

```

# Predict

y_pred = model.predict(X_test)

# Accuracy

print("Accuracy:", accuracy_score(y_test, y_pred))

# Classification Report

print("\nClassification Report:")

print(classification_report(y_test, y_pred))

```

OUTPUT:

Accuracy: 1.0

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```

... Accuracy: 1.0

Classification Report:
              precision    recall  f1-score   support

     0       1.00      1.00      1.00        10
     1       1.00      1.00      1.00         9
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 accuracy          1.00          30
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 weighted avg          1.00          30

```